ACS Catalysis

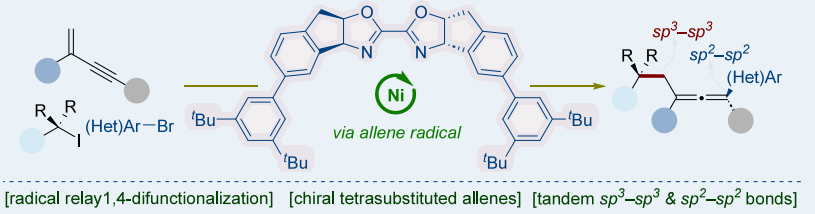
pubs.acs.org/acscatalysis Research Article

Enantio- and Regioselective Ni-Catalyzed Radical Relay 1,4-Arylalkylation of 1,3-Enynes to Access Chiral Tetrasubstituted Allenes

Jinze Du, Anyao Liu, Wen Zhang, Shan Luo, Hequan Yao,* and Yuli He*

Cite This: ACS Catal. 2024, 14, 13940−13946 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information



ABSTRACT: The enantio- and regioselective reductive 1,4-arylalkylation of 1,3-enynes has been successfully accomplished using a nickel catalyst in conjunction with a chiral bulk bis-oxazoline ligand. This method facilitates the efficient synthesis of a wide variety of structurally diverse, enantioenriched tetrasubstituted allenes, which are key structural motifs in numerous pharmacologically active compounds and natural products. The reaction proceeds under mild conditions and is operationally straightforward. Mechanistic investigations indicate that the reaction follows a radical pathway.

KEYWORDS: *nickel catalysis, radical relay, chiral tetrasubstituted allenes, asymmetric reductive coupling, 1,4-arylalkylation*

Downloaded via FUDAN UNIV on February 5, 2025 at 04:13:14 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

INTRODUCTION

In recent years, nickel catalysis has become pivotal in diverse transformations due to its ability to accommodate various oxidation states and engage in both one- and two-electron pathways.[1] This versatility has revolutionized cross-coupling and reductive coupling reactions,[2] addressing challenges posed by other transition metals in the d^{10} series. With the strategic selection of chiral ligands, nickel catalysis has enabled extensive exploration of enantioselective 1,2-difunctionalization of unsaturated hydrocarbons, facilitating the synthesis of valuable chiral molecules.[3] This is particularly significant given the pivotal role of enantioenriched sp^3-hybridized carbon centers in organic synthesis (Figure 1, left).[4] However, the enantioselective 1,4-difunctionalization of 1,3-enynes to access sp^2-hybridized carbon centers through nickel catalysis remains an untapped area of research (Figure 1, right).[5]

1,3-Enynes, which feature conjugated double and triple bonds, are versatile synthons in organic synthesis. Recently, the radical-based 1,4-functionalization of 1,3-enynes, utilizing allenyl or propargylic radicals, has emerged as an area of significant research interest, demonstrating potential for novel synthesis pathways and product diversification.[6] Despite this interest, achieving asymmetric 1,4-functionalization through radical pathways remains a formidable challenge,[7] yet it remains crucial due to its role in synthesizing chiral allenes,[8] which are not only critical structural elements in natural products and bioactive molecules but also serve as dynamic synthons and catalysts, significantly enhancing the scope of

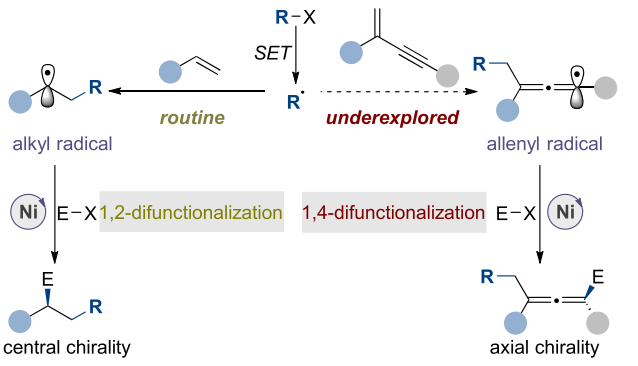


Figure 1. Enantioselective Ni-catalyzed cross-electrophile difunctionalization of alkenes.

organic synthesis (Figure 2a).[9] For instance, Bao[7a] and Liu[7b] independently described a Cu-catalyzed enantioselective process for synthesizing chiral allenes via the radical 1,4-dicarbylation of 1,3-enynes (Figure 2b up). Subsequently, the Wang group[7c] innovated with a dual photoredox and

Received: July 22, 2024
Revised: August 30, 2024
Accepted: August 30, 2024
Published: September 5, 2024



© 2024 American Chemical Society

https://doi.org/10.1021/acscatal.4c04371
ACS Catal. 2024, 14, 13940−13946

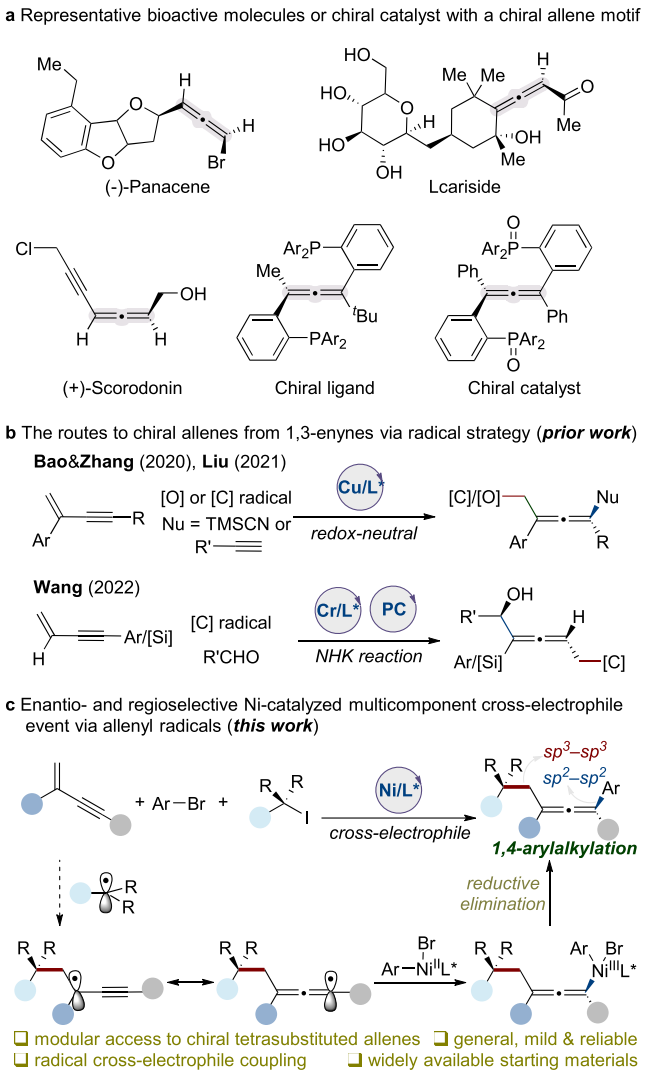


Figure 2. Design plan: enantio- and regioselective nickel-catalyzed reductive 1,4-arylalkylation of 1,3-enynes access to chiral tetrasubstituted allenes.

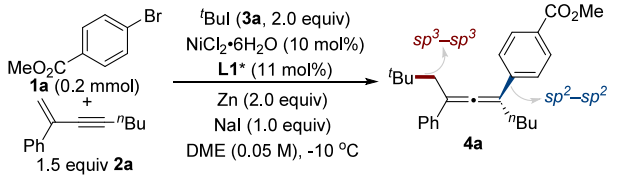
chromium catalytic system to achieve the synthesis of trisubstituted chiral α-allenols with notable enantio- and diastereoselectivities, facilitated by a propargylic radical intermediate (Figure 2b down). These radical-driven processes operate under mild conditions and enable the formation of densely functionalized complexes through a multicomponent approach, thus broadening the scope for modifying 1,3-enynes. The continued exploration of new radical-involved reaction mechanisms is anticipated to enhance the efficient synthesis of valuable chiral allenes, suggesting a promising direction for future research on organic synthesis.

In response to this challenge, we investigated the feasibility of a nickel-catalyzed 1,4-dicarbylation of 1,3-enynes to facilitate multicomponent cross-electrophile couplings for the synthesis of enantioenriched tetrasubstituted allenes. Herein, we report the successful development of a tandem nickel-catalyzed enantioselective $C(sp^3)-C(sp^3)$ and $C(sp^2)-C(sp^2)$ cross-coupling reaction between aryl bromides, unactivated alkyl halides, and enyne congeners (Figure 2c). This protocol, characterized by its simplicity and broad substrate scope, offers a complementary approach for preparing chiral tetrasubstituted allenes from simple precursors with precise control over selectivity at both alkene and allene double bond positions.

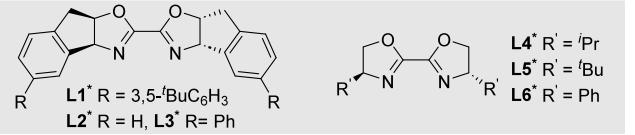
RESULTS AND DISCUSSION

Our investigations commenced with an examination of the reaction involving methyl 4-bromobenzoate (**1a**) with oct-1-en-3-yn-2-ylbenzene (**1a**) and 2-iodo-2-methylpropane (**3a**). After rigorous experimentation, we discovered that employing a combination of $NiCl_2{\cdot}6H_2O$, chiral bulk bis-oxazoline ligand (**L1***),[10] Zn as a reductant in DME at −10 °C delivered **4a** in 82% isolated yield with a single regioisomer and excellent enantioselectivity (92% ee) (Table 1, entry 1). The choice of

Table 1. Optimization of the Reaction Conditions



entry	deviation from standard conditions	yield of **4a** (%)[a]	*ee* (%)[b]
1	***none***	***79 (82)***	***92***
2	**L2*** instead of **L1***	75	78
3	**L3*** instead of **L1***	68	87
4	**L4*** instead of **L1***	15	56
5	**L5*** instead of **L1***	NR	ND
6	**L6*** instead of **L1***	5	ND
7	Mn instead of Zn	trace	ND
8	*w/o* $NiCl_2{\cdot}6H_2O$, **L1*** or NaI	NR	ND
9	ArI instead of ArBr	79	91
10	5 mol% [Ni], 6 mol% **L1***	57	91
11	rt instead of -10 °C	54	88
12	THF instead of DME	77	87



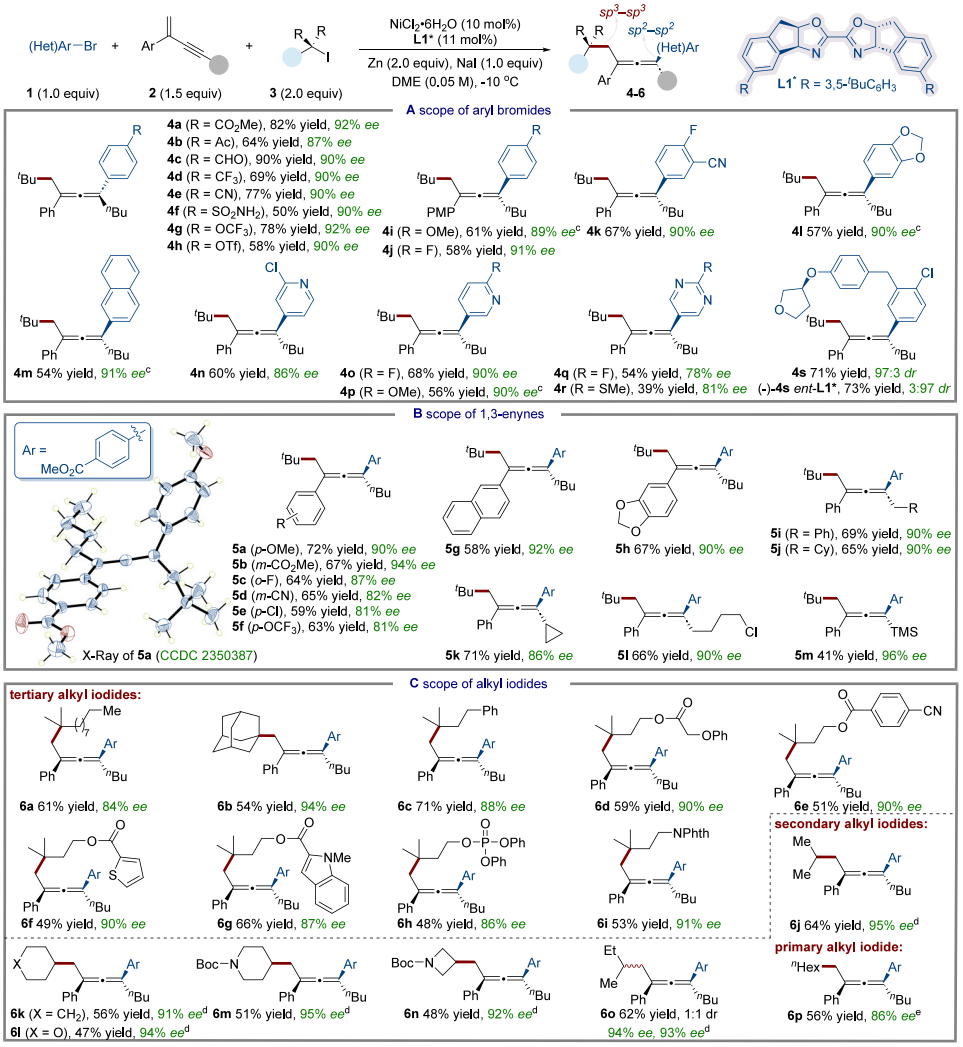
[a]Yields determined by GC using n-dodecane as internal standard, the yield in parentheses is the isolated yield. [b]Enantioselectivity was determined by chiral HPLC analysis. DME = 1,2-dimethoxyethane. THF = tetrahydrofuran. NR = no reaction. ND = not determined.

ligands significantly influenced the reaction outcome. Notably, other ligands (**L4*−L6***), known for their efficacy in asymmetric reductive difunctionalization of alkenes and cross-couplings,[4d, 4g] produced **4a** in low yields and ee values (entries 4−6). Conversely, employing a sterically hindered aryl substituent ligand (**L3***) resulted **4a** in 87% ee (entry 3 vs entry 2). Interestingly, we observed a detrimental effect on reactivity when Mn was used as a reductant or in the absence of NaI (entries 7−8). Specifically, similar yields and ee values were obtained for methyl 4-iodobenzoate (ArI) (entry 9). As illustrated in entries 10−12, lower catalyst loading, higher temperatures, and alternative solvents such as THF did not yield further improvements in the results.

With optimal conditions established, our attention turned to systematically exploring the preparative scope of the nickel-

https://doi.org/10.1021/acscatal.4c04371
ACS Catal. 2024, 14, 13940−13946

Table 2. Scope of Nickel-Catalyzed Enantio- and Regioselective Reductive 1,4-Arylalkylation of 1,3-Enynes[a,b]

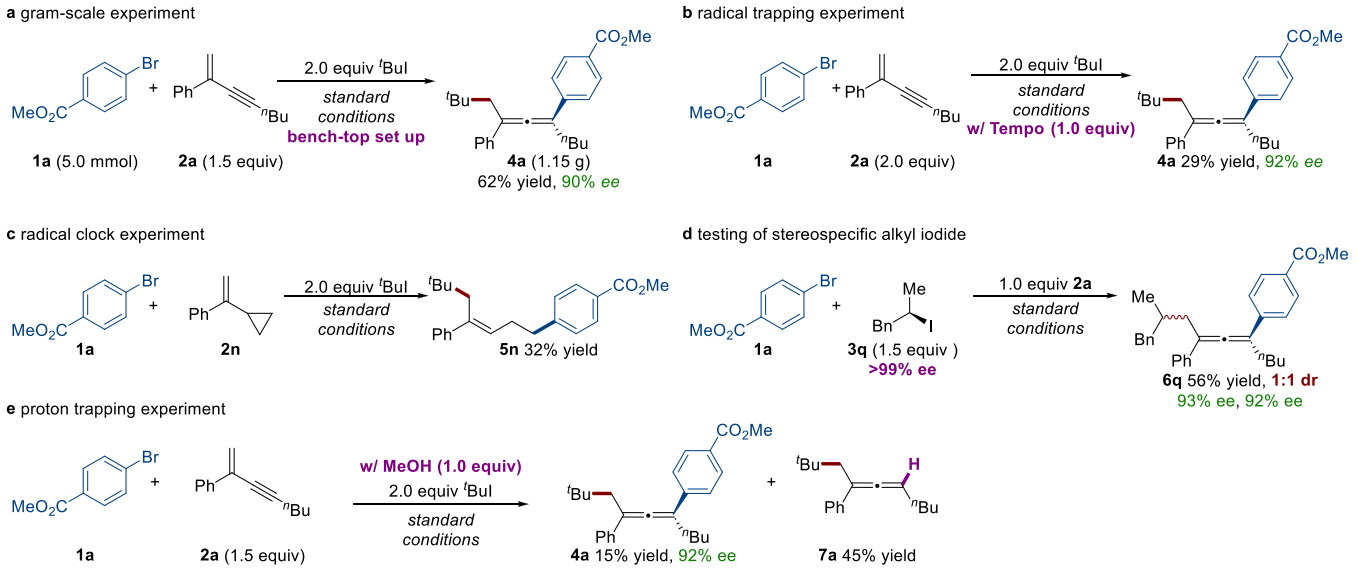


[a]Yield under each product refers to the isolated yield of purified product (0.20 mmol scale, average of two runs). [b]Enantioselectivities were determined by chiral HPLC analysis. [c](Het)Ar−I instead of (Het)Ar−Br. [d]**2** (0.20 mmol, 1.0 equiv), $NiCl_2$·dme (10 mol %), KI (1.5 equiv), and 0 °C were used. [e]NiI_2 (10 mol %), and THF were used.

catalyzed enantio- and regioselective reductive 1,4-arylalkylation of 1,3-enynes. The results tabulated in Table 2 underscore the remarkable versatility of our protocol across a diverse array of aryl bromides, alkyl iodides, and 1,3-enyne substrates. Notably, regardless of whether electron-deficient or electron-rich bromoarenes were utilized as coupling partners, consistently good yields and excellent ee values were achieved. An extensive range of functional groups, encompassing esters (**4a**), ketones (**4b**), aldehydes (**4c**), nitriles (**4e**, **4k**), sulfonamides (**4f**), and sulfur ethers (**4r**), were adeptly accommodated within the reaction framework. Importantly, the compatibility of our methodology with functionalities conducive to Ni-catalyzed transformations, such as aryl fluorides (**4j**, **4k**, **4o**, **4q**), aryl chlorides (**4n**, **4s**), and aryl triflates (**4h**), was demonstrated, ensuring avenues for further diversification through conventional cross-coupling methodologies remained open.[1] Heterocyclic motifs, including pyridines (**4n**−**4p**) and pyrimidines (**4q**−**4r**), emerged as effective coupling partners, further broadening the synthetic utility of our approach. Table 2B showcases the successful application of our methodology to a plethora of 1,3-enynes bearing both aromatic and aliphatic termini. Particularly noteworthy was the observation that enynes featuring electron-donating substituents on the arene moiety displayed enhanced enantioselectivity profiles (**5a**, **5g** vs **5b**−**5f**, **5h**). Crucially, the X-ray crystallographic analysis of product **5a** provided unambiguous evidence regarding the stereochemical outcome, elucidating the configuration of the difunctionalized arene with an *R*-configuration allene. Furthermore, enynes bearing ortho-substituted arenes (**5c**) were smoothly accommodated within the reaction manifold. In terms of 1,3-enynes possessing substituents at the 4-position, those featuring benzyl or purely aliphatic functional groups (**5j**−**5k**) were afforded in good yields with exemplary enantioselectivity. Notably, diverse functionalities, such as cyclopropyl (**5k**), chloride (**5l**), or silyl (**5m**) moieties, positioned at various distances from the allenyl radicals, were well-tolerated, furnishing chiral allenes with up to 96% ee. Our reaction exhibited consistent applicability across various unactivated alkyl iodides bearing different substituents (Table 2C). Notably, tertiary (**6a**−**6i**), secondary (**6j**−**6o**) as well as primary alkyl iodides (**6p**) displayed similar outcomes, affirming the robustness of our approach. The

https://doi.org/10.1021/acscatal.4c04371
ACS Catal. 2024, 14, 13940−13946

Scheme 1. Gram-Scale, Preliminary Mechanistic Studies



enantioselectivity of the reaction was underscored by its compatibility with diverse functional groups such as esters (**6d**−**6g**), phosphonates (**6h**), amides (**6i**), carbamates (**6m**, **6n**), nitriles (**6e**), thiophene (**6f**), and indole (**6g**), among others. Nonetheless, the stereochemical outcome was less controllable when unsymmetrical racemic secondary alkyl iodide substrates (**3o**), posing a challenge for precise stereocenter manipulation within this protocol.

The synthetic utility of our protocol is further highlighted in Scheme 1a. As shown, a 5 mmol scale enantioselective reductive 1,4-arylalkylation was performed successfully. Our investigation proceeded to elucidate the reaction mechanism through a series of control experiments. Introduction of the radical scavenger TEMPO (2,2,6,6-tetramethylpiperidinooxy) into the reaction mixture under the standard conditions resulted in a diminished yield. This reduction was accompanied by the conspicuous formation of radical-trapped products, confirmed by high-resolution mass spectrometry (refer to 6.1 in the Supporting Information) (Scheme 1b). To interrogate the reaction intermediates, a radical clock experiment was conducted using substrate **2n**. The ensuing reaction furnished the ring-opening product **5n** in 32% yield, providing compelling evidence for the involvement of alkyl radicals (Scheme 1c). Moreover, upon subjecting the stereospecific substrate **3q** to standard conditions, the resulting chiral allene **6q** was formed in a racemic manner (1:1 dr), thereby suggesting that the generation of secondary alkyl radical species induced racemization. In the presence of MeOH (1.0 equiv), the reaction yielded the desired product **4a** alongside the protonated byproduct **7a** (Scheme 1e). Notably, this outcome suggests the occurrence of a propargyl or allenyl radical intermediate prior to recombination with ArNi(II)Br. The reaction exhibits first-order dependence on nickel, ligand, and aryl bromide, underscoring their critical roles in the process. In contrast, the zero-order dependence on alkene, alkyl iodide, zinc, and sodium iodide indicates that these components do not substantially affect the reaction rate. Consequently, it is likely that the oxidative addition of aryl bromide is the rate-determining step (Scheme 1f). Additionally, drawing on the significant insights from Fu,[11a] Chu,[11b−d]

https://doi.org/10.1021/acscatal.4c04371
ACS Catal. 2024, 14, 13940−13946

and Bao,[7a] we anticipate that the $\pi-\pi$ interaction between the allenyl radical and the chiral aryl-substituted BiOx-ligated nickel complex will help control the stereoselectivity during the radical capture step.

In light of these mechanistic insights and prior research, a proposed catalytic cycle is outlined in Figure 3. We postulated

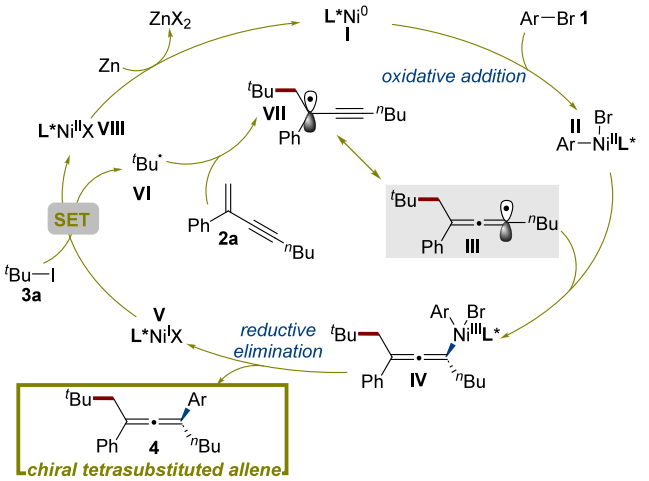


Figure 3. Proposed mechanistic pathway.

that a suitable L*Ni(0) catalyst could facilitate oxidative addition at the Ar−Br site, leading to the formation of the ArNi(II)Br **II** intermediate. Simultaneously, Ni(I) species generated in the reaction mixture could activate the alkyl iodide,[12] generating an alkyl radical **VI** which then adds onto the C = C bond to form the propargyl radical **VII** and allenyl radical **III**. Subsequently, ArNi(II)Br **II** can react with intermediate **III** to yield pivotal Ni(III) species **IV**. Reductive elimination of **IV** would furnish the desired difunctionalized framework (**4**) along with a Ni(I) intermediate (**V**) that would ultimately regenerate the catalytic L*Ni(0) species upon single-electron transfer (SET) with an alkyl halide (**3a**) and Zn as the terminal reductant.

CONCLUSIONS

In summary, we have developed a de novo platform that facilitates the synthesis of 1,4-difunctionalized allenes with outstanding enantioselectivities, catalyzed by nickel utilizing a novel chiral bulk bis-oxazoline ligand. This methodology stands out for its broad substrate applicability and mild reaction conditions, thereby providing a complementary approach for achieving enantioselective reductive nickel-catalyzed multicomponent difunctionalization of alkenes. Preliminary mechanistic investigations indicated the involvement of a radical tandem process in the reaction. Ongoing efforts are focused on expanding the scope of asymmetric cross-electrophile 1,n-difunctionalization of alkenes, while simultaneously progressing toward a deeper understanding of the stereoselectivity mechanisms through DFT studies.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.4c04371.

Experimental procedures and characterization data for all compounds (PDF)

Crystallographic data of **5a** (CIF)

AUTHOR INFORMATION

Corresponding Authors

Hequan Yao − *State Key Laboratory of Natural Medicines (SKLNM) and Department of Medicinal Chemistry, School of Pharmacy, China Pharmaceutical University, Nanjing 211198, P. R. China*; Email: hyao@cpu.edu.cn, cpuhyao@126.com

Yuli He − *State Key Laboratory of Natural Medicines (SKLNM) and Department of Medicinal Chemistry, School of Pharmacy, China Pharmaceutical University, Nanjing 211198, P. R. China*; orcid.org/0000-0003-0279-5647; Email: yulihe@cpu.edu.cn

Authors

Jinze Du − *State Key Laboratory of Natural Medicines (SKLNM) and Department of Medicinal Chemistry, School of Pharmacy, China Pharmaceutical University, Nanjing 211198, P. R. China*

Anyao Liu − *State Key Laboratory of Natural Medicines (SKLNM) and Department of Medicinal Chemistry, School of Pharmacy, China Pharmaceutical University, Nanjing 211198, P. R. China*

Wen Zhang − *State Key Laboratory of Natural Medicines (SKLNM) and Department of Medicinal Chemistry, School of Pharmacy, China Pharmaceutical University, Nanjing 211198, P. R. China*

Shan Luo − *State Key Laboratory of Natural Medicines (SKLNM) and Department of Medicinal Chemistry, School of Pharmacy, China Pharmaceutical University, Nanjing 211198, P. R. China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acscatal.4c04371

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

The authors acknowledge generous financial support from the National Natural Science Foundation of China (22371298, 22371299, and 22071267), and the Foundation of The Open Project of State Key Laboratory of Natural Medicines (SKLNMZZ202211). We thank Prof. Shaolin Zhu (NJU), Prof. Aijun Lin, and Prof. Shang Gao (CPU) for inspiring discussions.

REFERENCES

(1) For selected reviews on nickel catalysis, see: (a) Hu, X. Nickel-Catalyzed Cross Coupling of Non-Activated Alkyl Halides: A Mechanistic Perspective. *Chem. Sci.* **2011**, *2*, 1867−1886. (b) Montgomery, J. Organonickel Chemistry. In *Organometallics in Synthesis*; Lipshutz, B. H., Ed.; John Wiley & Sons, Inc.: Hoboken, 2013; pp 319−428. (c) Tasker, S. Z.; Standley, E. A.; Jamison, T. F. Recent Advances in Homogeneous Nickel Catalysis. *Nature* **2014**, *509*, 299−309. (d) Ananikov, V. P. Nickel: The "Spirited Horse" of Transition Metal Catalysis. *ACS Catal.* **2015**, *5*, 1964−1971. (e) Fu, G. C. Transition-Metal Catalysis of Nucleophilic Substitution Reactions: a Radical Alternative to S_N1 and S_N2 Processes. *ACS Cent. Sci.* **2017**, *3*, 692−700. (f) Ogoshi, S. *Nickel Catalysis in Organic Synthesis*; Wiley-VCH: Weinheim, 2020. (g) Dawson, G. A.; Spielvogel, E. H.; Diao, T. Nickel-Catalyzed Radical Mechanisms: Informing Cross-Coupling for Synthesizing Non-Canonical Biomolecules. *Acc. Chem. Res.* **2023**, *56*, 3640−3653. (h) Li, L.-J.; He, Y.; Yang, Y.; Guo, J.; Lu, Z.; Wang, C.;

Zhu, S.; Zhu, S.-F. Recent Advances in Mn, Fe, Co, and Ni-Catalyzed Organic Reactions. *CCS Chem.* **2024**, *6*, 537−584.

(2) For selected reviews on nickel-catalyzed reductive cross-coupling, see: (a) Everson, D. A.; Weix, D. J. Cross-Electrophile Coupling: Principles of Reactivity and Selectivity. *J. Org. Chem.* **2014**, *79*, 4793−4798. (b) Moragas, T.; Correa, A.; Martin, R. Metal-Catalyzed Reductive Coupling Reactions of Organic Halides with Carbonyl-Type Compounds. *Chem. − Eur. J.* **2014**, *20*, 8242−8258. (c) Weix, D. J. Methods and Mechanisms for Cross-Electrophile Coupling of Csp^2 Halides with Alkyl Electrophiles. *Acc. Chem. Res.* **2015**, *48*, 1767−1775. (d) Jackson, E. P.; Malik, H. A.; Sormunen, G. J.; Baxt, R. D.; Liu, P.; Wang, H.; Shareef, A.-R.; Montgomery, J. Mechanistic Basis for Regioselection and Regiodivergence in Nickel-Catalyzed Reductive Couplings. *Acc. Chem. Res.* **2015**, *48*, 1736−1745. (e) Wang, X.; Dai, Y.; Gong, H. Nickel-Catalyzed Reductive Couplings. *Top. Curr. Chem.* **2016**, *374*, 43−72. (f) Diccianni, J. B.; Diao, T. Mechanisms of Nickel-Catalyzed Cross-Coupling Reactions. *Trends Chem.* **2019**, *1*, 830−844. (g) Poremba, K. E.; Dibrell, S. E.; Reisman, S. E. Nickel-Catalyzed Enantioselective Reductive Cross-Coupling Reactions. *ACS Catal.* **2020**, *10*, 8237−8246. (h) Charboneau, D. J.; Hazari, N.; Huang, H.; Uehling, M. R.; Zultanski, S. L. Homogeneous Organic Electron Donors in Nickel-Catalyzed Reductive Transformations. *J. Org. Chem.* **2022**, *87*, 7589−7609. (i) Huang, H.; Alvarez-Hernandez, J. L.; Hazari, N.; Mercado, B. Q.; Uehling, M. R. Effect of 6,6′-Substituents on Bipyridine-Ligated Ni Catalysts for Cross-Electrophile Coupling. *ACS Catal.* **2024**, *14*, 6897−6914. (j) Chen, L.-M.; Reisman, S. E. Enantioselective $C(sp^2)-C(sp^3)$ Bond Construction by Ni Catalysis. *Acc. Chem. Res.* **2024**, *57*, 751−762.

(3) For selected reviews on nickel-catalyzed difunctionalization of alkenes, see: (a) Badir, S. O.; Molander, G. A. Developments in Photoredox/Nickel Dual-Catalyzed 1,2-Difunctionalizations. *Chem.* **2020**, *6*, 1327−1339. (b) Derosa, J.; Apolinar, O.; Kang, T.; Tran, V. T.; Engle, K. M. Recent Developments in Nickel-Catalyzed Intermolecular Dicarbofunctionalization of Alkenes. *Chem. Sci.* **2020**, *11*, 4287−4296. (c) Zhu, S.; Zhao, X.; Li, H.; Chu, L. Catalytic Three-Component Dicarbofunctionalization Reactions Involving Radical Capture by Nickel. *Chem. Soc. Rev.* **2021**, *50*, 10836−10856. (d) Luo, Y.-C.; Xu, C.; Zhang, X. Nickel-Catalyzed Dicarbofunctionalization of Alkenes. *Chin. J. Chem.* **2020**, *38*, 1371−1394. (e) Qi, X.; Diao, T. Nickel-Catalyzed Dicarbofunctionalization of Alkenes. *ACS Catal.* **2020**, *10*, 8542−8556. (f) Wickham, L.; Giri, R. Transition Metal (Ni, Cu, Pd)-Catalyzed Alkene Dicarbofunctionalization Reactions. *Acc. Chem. Res.* **2021**, *54*, 3415−3437. (g) Ping, Y.; Song, H.; Kong, W. Recent Advances in Ni-Catalyzed Asymmetric Reductive Difunctionalization of Alkenes. *Chin. J. Org. Chem.* **2022**, *42*, 3302−3321. (h) Lee, B. C.; Liu, C.-F.; Lin, L. Q. H.; Yap, K. Z.; Song, N.; Ko, C. H. M.; Chan, P. H.; Koh, M. J. N-Heterocyclic Carbenes as Privileged Ligands for Nickel-Catalysed Alkene Functionalization. *Chem. Soc. Rev.* **2023**, *52*, 2946−2991. (i) Kang, T.; Apolinar, O.; Engle, K. M. Ni- and Pd-Catalyzed Enantioselective 1,2-Dicarbofunctionalization of Alkenes. *Synthesis* **2024**, *56*, 1−15.

(4) For select examples of Ni-catalyzed (reductive) asymmetric 1,2-difunctionalization, see: (a) Cong, H.; Fu, G. C. Catalytic Enantioselective Cyclization/Cross-Coupling with Alkyl Electrophiles. *J. Am. Chem. Soc.* **2014**, *136*, 3788−3791. (b) Wang, K.; Ding, Z.; Zhou, Z.; Kong, W. Ni-Catalyzed Enantioselective Reductive Diarylation of Activated Alkenes by Domino Cyclization/Cross-Coupling. *J. Am. Chem. Soc.* **2018**, *140*, 12364−12368. (c) Tian, Z.; Qiao, J.; Xu, G.; Pang, X.; Qi, L.; Ma, W.; Zhao, Z.; Duan, J.; Du, Y.; Su, P.; Liu, X.; Shu, X. Highly Enantioselective Cross-Electrophile Aryl-Alkenylation of Unactivated Alkenes. *J. Am. Chem. Soc.* **2019**, *141*, 7637−7643. (d) Anthony, D.; Lin, Q.; Baudet, J.; Diao, T. Nickel-Catalyzed Asymmetric Reductive Diarylation of Vinylarenes. *Angew. Chem., Int. Ed.* **2019**, *58*, 3198−3202. (e) Jin, Y.; Wang, C. Nickel-Catalyzed Asymmetric Reductive Arylalkylation of Unactivated Alkenes. *Angew. Chem., Int. Ed.* **2019**, *58*, 6722−7626. (f) Tu, H.; Wang, F.; Huo, L.; Li, Y.; Zhu, S.; Zhao, X.; Li, H.; Qing, F.; Chu, L. Enantioselective Three-Component Fluoroalkylarylation of Unactivated Olefins Through Nickel-Catalyzed Cross-Electrophile Coupling. *J. Am. Chem. Soc.* **2020**, *142*, 9604−9611. (g) Wei, X.; Shu, W.; Andrés, G. D.; Estíbaliz, M.; Nevado, C. Asymmetric Ni-Catalyzed Radical Relayed Reductive Coupling. *J. Am. Chem. Soc.* **2020**, *142*, 13515−13522. (h) Wu, X.; Qu, J.; Chen, Y. Quinim: a New Ligand Scaffold Enables Nickel-Catalyzed Enantioselective Synthesis of α-Alkylated γ-Lactam. *J. Am. Chem. Soc.* **2020**, *142*, 15654−15660. (i) Qian, P.; Guan, H.; Wang, Y.; Lu, Q.; Zhang, Q. F.; Xiong, D.; Walsh, P.; Mao, J. Catalytic Enantioselective Reductive Domino Alkyl Arylation of Acrylates via Nickel/Photoredox Catalysis. *Nat. Commun.* **2021**, *12*, 6613−6622. (j) Li, Y.-Z.; Rao, N.; An, L.; Wan, X.-L.; Zhang, Y.; Zhang, X. Enantioselective Nickel-Catalyzed Dicarbofunctionalization of 3,3,3-Trifluoropropene. *Nat. Commun.* **2022**, *13*, 5539−5548. (k) Apolinar, O.; Kang, T.; Alturaifi, T. M.; Bedekar, P. G.; Rubel, C. Z.; Derosa, J.; Sanchez, B. B.; Wong, Q. N.; Sturgell, E. J.; Chen, J. S.; Wisniewski, S. R.; Liu, P.; Engle, K. M. Three-Component Asymmetric Ni-Catalyzed 1,2-Dicarbofunctionalization of Unactivated Alkenes via Stereoselective Migratory Insertion. *J. Am. Chem. Soc.* **2022**, *144*, 19337−19343. (l) Liu, C.-F.; Wang, Z.-C.; Luo, X.; Lu, J.; Ko, C. H. M.; Shi, S.-L.; Koh, M. J. Synthesis of Tri- and Tetrasubstituted Stereocentres by Nickel-Catalysed Enantioselective Olefin Cross-Couplings. *Nat. Catal.* **2022**, *5*, 934−942. (m) Dong, Z.; Tang, Q.; Xu, C.; Chen, L.; Ji, H.; Zhou, S.; Song, L.; Chen, L.-A. Directed Asymmetric Nickel-Catalyzed Reductive 1,2-Diarylation of Electronically Unactivated Alkenes. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202218286. (n) Wang, Y.-Z.; Sun, B.; Zhu, X.-Y.; Gu, Y.-C.; Ma, C.; Mei, T.-S. Enantioselective Reductive Cross-Couplings of Olefins by Merging Electrochemistry with Nickel Catalysis. *J. Am. Chem. Soc.* **2023**, *145*, 23910−23917. (o) Hu, X.; Cheng-Sánchez, I.; Kong, W.; Molander, G. A.; Nevado, C. Nickel-Catalysed Enantioselective Alkene Dicarbofunctionalization Enabled by Photochemical Aliphatic C−H Bond Activation. *Nat. Catal.* **2024**, *7*, 655−665.

(5) For selected racemic examples about nickel catalyzed 1,4-difunctionalization of 1,3-enynes, see: (a) Zhang, K.-F.; Bian, K.-J.; Li, C.; Sheng, J.; Li, Y.; Wang, X.-S. Nickel-Catalyzed Carbofluoroalkylation of 1,3-Enynes to Access Structurally Diverse Fluoroalkylated Allenes. *Angew. Chem., Int. Ed.* **2019**, *58*, 5069−5074. (b) Chen, Y.; Zhu, K.; Huang, Q.; Lu, Y. Regiodivergent Sulfonylarylation of 1,3-Enynes via Nickel/Photoredox Dual Catalysis. *Chem. Sci.* **2021**, *12*, 13564−13571. (c) Liu, W.; Liu, C.; Wang, M.; Kong, W. Modular Synthesis of Multifunctionalized CF_3-Allenes Through Selective Activation of Saturated Hydrocarbons. *ACS Catal.* **2022**, *12*, 10207−10221. (d) Cheng, C.; Lv, G.-F.; Wu, S.; Li, Y.; Li, J.-H. Nickel Photocatalyzed Reductive 1,4-Dicarbofunctionalization of 1,3-Enynes with N-Methylamines and Organohalides Enabled by Site-Selective $C(sp^3)-H$ Functionalization. *Org. Lett.* **2023**, *25*, 4236−4240. (e) Chen, H.; Zhu, C.; Yue, H.; Rueping, M. Groupn 14 Elements Hetero-Difunctionalizations via Nickel-Catalyzed Electroreductive Cross-Coupling. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202306498. (f) Zhu, C.; Chen, H.; Yue, H.; Rueping, M. Electrochemical Chemo- and Regioselective Arylalkylation, Dialkylation and Hydro(deutero)alkylation of 1,3-Enynes. *Nat. Synth.* **2023**, *2*, 1068−1081. (g) Zhang, G.; Tan, W.; Zhang, D.; Wang, K.; Gao, P.; Wang, S.; Liu, S.-L.; Chen, F. Regioselective Hydro(deutero)silylation of 1,3-Enynes Enabled by Photoredox/Nickel Dual Catalysis. *Org. Lett.* **2024**, *26*, 536−541. (h) Shan, Q.-C.; Zhao, Y.; Wang, S.-T.; Liu, H.-F.; Duan, X.-H.; Guo, L.-N. Nickel-Catalyzed Modular Four-Component 1,4-Alkylcarbonylation of 1,3-Enynes to Tetra-Substituted CF_3−Allenyl Ketones. *ACS Catal.* **2024**, *14*, 2144−2150. (i) Jeon, J. H.; Kim, G. H.; Lee, H. S.; Kim, D. H.; Lee, S.; Choe, W.; Jung, B.; Rohde, J.-U.; Hong, S. Y. Chemo- and Regioselective Nickel-Catalyzed Reductive 1,4-Alkylarylation of 1,3-Enynes Through an L_2NiAr Intermediate. *ACS Catal.* **2024**, *14*, 8996−9007.

(6) For selected reviews of the 1,4-Functionalizations from 1,3-enynes, see: (a) Terao, J.; Bando, F.; Kambe, N. Ni-Catalyzed Regioselective Three-Component Coupling of Alkyl Halides, Arylalkynes, or Enynes with R−M (M = MgX′, ZnX′). *Chem. Commun.* **2009**, 7336−7338. (b) Fu, L.; Greßies, S.; Chen, P.; Liu, G.

https://doi.org/10.1021/acscatal.4c04371
ACS Catal. 2024, 14, 13940−13946

Recent Advances and Perspectives in Transition Metal-Catalyzed 1,4-Functionalizations of Unactivated 1,3-Enynes for The Synthesis of Allenes. *Chin. J. Chem.* **2020**, *38*, 91−100. (c) Dherbassy, Q.; Manna, S.; Talbot, F. J. T.; Prasitwatcharakorn, W.; Perry, G. J. P.; Procter, D. J. Copper-Catalyzed Functionalization of Enynes. *Chem. Sci.* **2020**, *11*, 11380−11393. (d) Li, Y.; Bao, H. Radical Transformations for Allene Synthesis. *Chem. Sci.* **2022**, *13*, 8491−8506.

(7) (a) Zeng, Y.; Chiou, M.-F.; Zhu, X.; Cao, J.; Lv, D.; Jian, W.; Li, Y.; Zhang, X.; Bao, H. Copper-Catalyzed Enantioselective Radical 1,4-Difunctionalization of 1,3-Enynes. *J. Am. Chem. Soc.* **2020**, *142*, 18014−18021. (b) Dong, X.-Y.; Zhan, T.-Y.; Jiang, S.-P.; Liu, X.-D.; Ye, L.; Li, Z.-L.; Gu, Q.-S.; Liu, X.-Y. Copper-Catalyzed Asymmetric Coupling of Allenyl Radicals with Terminal Alkynes to Access Tetrasubstituted Allenes. *Angew. Chem., Int. Ed.* **2021**, *60*, 2160−2164. (c) Zhang, F.-H.; Guo, X.; Zeng, X.; Wang, Z. Asymmetric 1,4-Functionalization of 1,3-Enynes via Dual Photoredox and Chromium Catalysis. *Nat. Commun.* **2022**, *13*, 5036−5047.

(8) For selected examples of the synthesis of axially chiral allenes catalyzed by transition metals, see: (a) Wan, B.; Ma, S. Enantioselective Decarboxylative Amination: Synthesis of Axially Chiral Allenyl Amines. *Angew. Chem., Int. Ed.* **2013**, *52*, 441−445. (b) Huang, Y.; del Pozo, J.; Torker, S.; Hoveyda, A. H. Enantioselective Synthesis of Trisubstituted Allenyl-B(pin) Compounds by Phosphine-Cu-Catalyzed 1,3-Enyne Hydroboration. Insights Regarding Stereochemical Integrity of Cu-Allenyl Intermediates. *J. Am. Chem. Soc.* **2018**, *140*, 2643−2655. (c) Adamson, N. J.; Jeddi, H.; Malcolmson, S. J. Preparation of Chiral Allenes through Pd-Catalyzed Intermolecular Hydroamination of Conjugated Enynes: Enantioselective Synthesis Enabled by Catalyst Design. *J. Am. Chem. Soc.* **2019**, *141*, 8574−8583. (d) Bayeh-Romero, L.; Buchwald, S. L. Copper Hydride Catalyzed Enantioselective Synthesis of Axially Chiral 1,3-Disubstituted Allenes. *J. Am. Chem. Soc.* **2019**, *141*, 13788−13794. (e) Zheng, W.-F.; Zhang, W.; Huang, C.; Wu, P.; Qian, H.; Wang, L.; Guo, Y.-L.; Ma, S. Tetrasubstituted Allenes via The Palladium-Catalysed Kinetic Resolution of Propargylic Alcohols Using a Supporting Ligand. *Nat. Catal.* **2019**, *2*, 997−1005. (f) Zhu, C.; Chu, H.; Li, G.; Ma, S.; Zhang, J. Pd-Catalyzed Enantioselective Heck Reaction of Aryl Triflates and Alkynes. *J. Am. Chem. Soc.* **2019**, *141*, 19246−19251. (g) He, C.-Y.; Tan, Y.-X.; Wang, X.; Ding, R.; Wang, Y.-F.; Wang, F.; Gao, D.; Tian, P.; Lin, G.-Q. Copper(I)-Catalyzed Diastereo- and Enantioelective Construction of Optically Pure Exocyclic Allenes. *Nat. Commun.* **2020**, *11*, 4293. (h) Zhang, J.; Huo, X.; Xiao, J.; Zhao, L.; Ma, S.; Zhang, W. Enantio- and Diastereodivergent Construction of 1,3-Nonadjacent Stereocenters Bearing Axial and Central Chirality through Synergistic Pd/Cu Catalysis. *J. Am. Chem. Soc.* **2021**, *143*, 12622−12632. (i) O'Connor, T. J.; Mai, B. K.; Nafie, J.; Liu, P.; Toste, F. D. Generation of Axially Chiral Fluoroallenes through a Copper-Catalyzed Enantioselective beta-Fluoride Elimination. *J. Am. Chem. Soc.* **2021**, *143*, 13759−13768. (j) Shi, B.; Liu, J.-B.; Wang, Z.-T.; Wang, L.; Lan, Y.; Lu, L.-Q.; Xiao, W.-J. Synthesis of Chiral Endocyclic Allenes by Palladium-Catalyzed Asymmetric Annulation Followed by Cope Rearrangement. *Angew. Chem., Int. Ed.* **2022**, *61*, No. e202117215. (k) Xu, X.; Wang, M.; Peng, L.; Guo, C. Nickel-Catalyzed Asymmetric Propargylation for The Synthesis of Axially Chiral 1,3-Disubstituted Allenes. *J. Am. Chem. Soc.* **2022**, *144*, 21022−21029. (l) Hossain, A.; Anderson, R. L.; Zhang, C. S.; Chen, P.-J.; Fu, G. C. Nickel-Catalyzed Enantioconvergent and Diastereoselective Allenylation of Alkyl Electrophiles: Simultaneous Control of Central and Axial Chirality. *J. Am. Chem. Soc.* **2024**, *146*, 7173−7177. (m) Liu, G.; Yang, X.; Gu, P.; Wang, M.; Zhang, X.; Dong, X.-Q. Challenging Task of Ni-Catalyzed Highly Regio-/Enantioselective Semihydrogenation of Racemic Tetrasubstituted Allenes via a Kinetic Resolution Process. *J. Am. Chem. Soc.* **2024**, *146*, 7419−7430. (n) Song, C.; Bai, X.; Li, B.; Dang, Y.; Yu, S. Photoexcited Palladium-Catalyzed Deracemization of Allenes. *J. Am. Chem. Soc.* **2024**, *146*, 21137−21146.

(9) (a) Hoffmann-Röder, A.; Krause, N. Synthesis and Properties of Allenic Natural Products and Pharmaceuticals. *Angew. Chem., Int. Ed.* **2004**, *43*, 1196−1216. (b) Ma, S. Some Typical Advances in The Synthetic Applications of Allenes. *Chem. Rev.* **2005**, *105*, 2829−2872. (c) Yu, S.; Ma, S. Allenes in Catalytic Asymmetric Synthesis and Natural Product Syntheses. *Angew. Chem., Int. Ed.* **2012**, *51*, 3074−3112. (d) Rivera-Fuentes, P.; Diederich, F. Allenes in Molecular Materials. *Angew. Chem., Int. Ed.* **2012**, *51*, 2818−2828. (e) Zhao, J.; Xu, Y. *Axially Chiral Compounds: Asymmetric Synthesis and Applications*; Tan, B., Ed.; Wiley-VCH: 2021; Chapter 6.

(10) Meng, L.; Yang, J.; Duan, M.; Wang, Y.; Zhu, S. Facile Synthesis of Chiral Arylamines, Alkylamines and Amides by Enantioselective NiH-Catalyzed Hydroamination. *Angew. Chem., Int. Ed.* **2021**, *60*, 23584−23589.

(11) (a) Yin, H.; Fu, G. C. Mechanistic Investigation of Enantioconvergent Kumada Reactions of Racemic α-Bromoketones Catalyzed by a Nickel/Bis(oxazoline) Complex. *J. Am. Chem. Soc.* **2019**, *141*, 15433−15440. (b) Guo, L.; Yuan, M.; Wang, Y. Z.; Zhu, S.; Gutierrez, O.; Chu, L. General Method for Enantioselective Three-Component Carboarylation of Alkenes Enabled by Visible-Light DualPhotoredox/Nickel Catalysis. *J. Am. Chem. Soc.* **2020**, *142*, 20390−20399. (c) Wang, F.; Pan, S.; Zhu, S.; Chu, L. Selective Three-Component Reductive Alkylalkenylation of Unbiased Alkenes via Carbonyl-Directed Nickel Catalysis. *ACS Catal.* **2022**, *12*, 9779−9789. (d) Li, X.; Yuan, M.; Chen, F.; Huang, Z.; Qing, F.-L.; Gutierrez, O.; Chu, L. Three-Component Enantioselective Alkenylation of Organophosphonates via Nickel Metallaphotoredox Catalysis. *Chem.* **2023**, *9*, 154−169.

(12) (a) Kehoe, R.; Mahadevan, M.; Manzoor, A.; McMurray, G.; Wienefeld, P.; Baird, M. C. Reactions of The Ni(0) Compound $Ni(PPh_3)_4$ with Unactivated Alkyl Halides: Oxidative Addition Reactions Involving Radical Processes and Nickel(I) Intermediates. *Organometallics* **2018**, *37*, 2450−2467. (b) Wang, X.; Ma, G.; Peng, Y.; Pitsch, C. E.; Moll, B. J.; Ly, T. D.; Wang, X.; Gong, H. Ni-Catalyzed Reductive Coupling of Electron-Rich Aryl Iodides with Tertiary Alkyl Halides. *J. Am. Chem. Soc.* **2018**, *140*, 14490−14497. (c) Diccianni, J. B.; Katigbak, J.; Hu, C.; Diao, T. Mechanistic Characterization of (Xantphos)Ni(I)-Mediated Alkyl Bromide Activation: Oxidative Addition, Electron Transfer, or Halogen-Atom Abstraction. *J. Am. Chem. Soc.* **2019**, *141*, 1788−1796.

https://doi.org/10.1021/acscatal.4c04371
ACS Catal. 2024, 14, 13940−13946